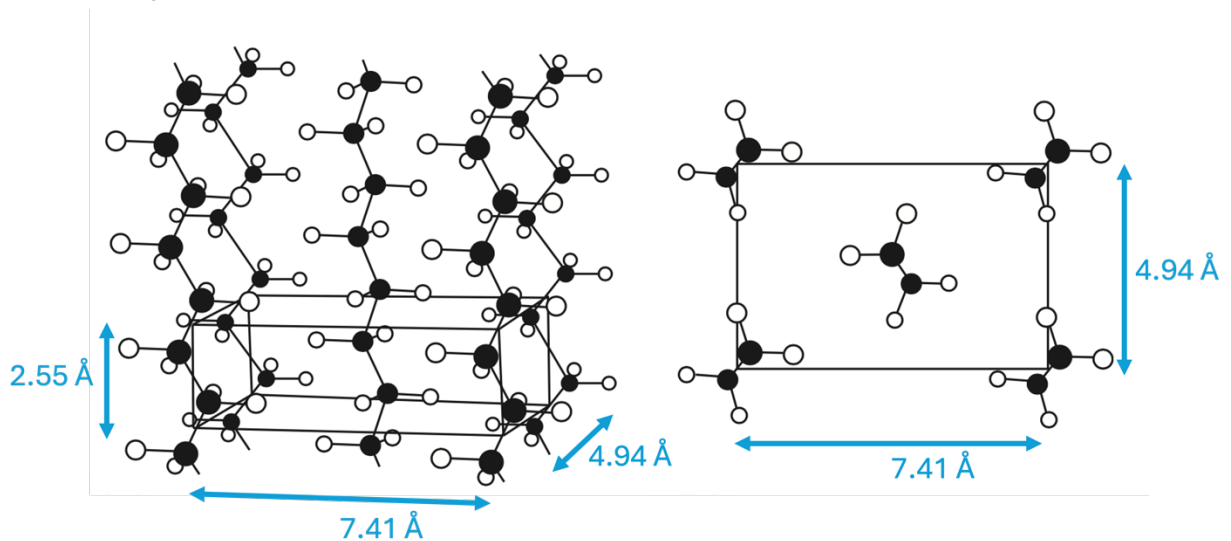


Be sure to carefully mark the area for the answers to each sub-question in Gradescope.

1. A sample of polylactic acid (PLA) has a bulk (or compression) modulus, K , of 293.8 MPa and a shear modulus, G , of 100.1 MPa.
 - a. Calculate the sample's Young's modulus, E , and its Poisson ratio, ν .
 - b. Based on its Young's modulus, is the sample more likely to be a glassy or semi-crystalline polymer?

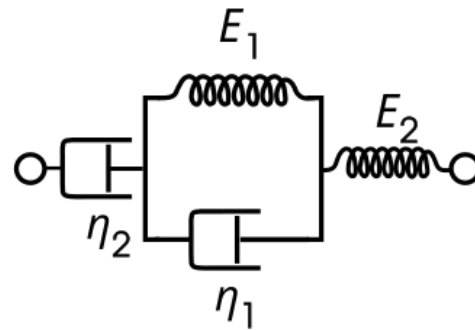
2. The crystal structure of polyethylene and its unit cell dimensions are given below. Assume that the length of a C-C bond is $1.54 \text{ \AA}$, and that a tetrahedral bond angle is 109.5° . The C-C bond force constant, k_l , is 436 Pa/m , and the bond angle force constant, k_θ , is $8.19 \times 10^{-19} \text{ Pa/m}$.



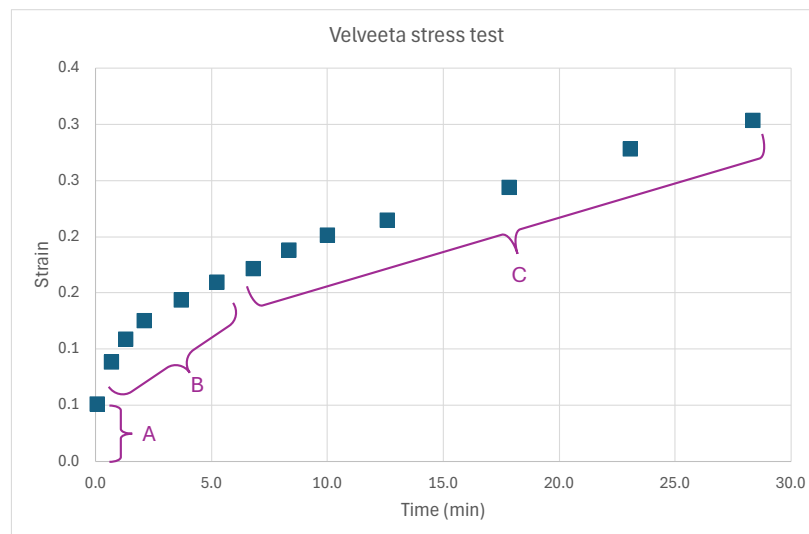
- a. Use Treolar's approach to calculate the elastic modulus of polyethylene when a tensile force is applied along the axis of the backbone.
- b. The elastic modulus of polyethylene at room temperature is actually 668 MPa. Analyze how close your estimate is, and explain any discrepancies.

3. You've been given a standard 32 oz block of Velveeta® cheese. Its dimensions are 6.5 cm x 7 cm x 21.5 cm. You apply a compression load of 4.2 N for 30 min. You measured the compression of the cheese block under load to obtain the following data:

Time (min)	Strain
0	0.051
0.7	0.089
1.3	0.109
2.1	0.125
3.7	0.144
5.2	0.160
6.8	0.172
8.3	0.188
10	0.202
12.6	0.215
17.8	0.244
23.1	0.278
28.3	0.304



- a. Assign the elements (E_1 , η_1 , E_2 , η_2) of the Burgers model to the regions in the plot below.



- b. Using the four-element model depicted above, estimate the values for each element (E_1 , η_1 , E_2 , η_2). The slope of region C is $6.099 \times 10^{-3} \text{ min}^{-1}$, and τ_1 is estimated to be 1.58 min.

- c. Sketch the strain vs time curve of the system as it is compressed and then is allowed to recover. Assign the elements during recovery as in (a).
 - d. After 30 minutes, you remove the compressive force. What is the strain of the block of cheese immediately after the compressive force has been removed? The last data point is at 28.3 minutes.
 - e. What is the final length of the block of cheese after it is allowed to recover for a very long time (i.e. irrecoverable creep)?
4. A biomedical device company is developing a new biodegradable surgical mesh using polyhydroxybutyrate (PHB). The viscoelastic properties of PHB are critical for this application, as the mesh must maintain its structural integrity during the initial healing period but gradually relax over time. The T_g of PHB is 5 °C, and the universal constants are: $C_1 = 17.44$, $C_2 = 51.6$ °C.
- a. Calculate the shift factor, a_T , at body temperature, 37 °C.
 - b. If a relaxation process takes 2 seconds at 37°C, how long would this process take at the storage temperature (14°C)?
 - c. At what temperature will the relaxation process take 380 seconds?